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EDA NYC Taxi Analysis – 2023

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1. Introduction

This analysis focuses on the 2023 NYC Yellow Taxi dataset to identify patterns in taxi demand, revenue trends, and passenger behavior. The goal is to derive insights that can help optimize taxi operations, improve efficiency, and enhance customer experience.

2. Data Sampling

The dataset contains millions of records, making it computationally expensive to process fully.

To handle this:

- Stratified sampling was performed based on **date and hour**
- Approximately **0.7% of data was sampled**
- Final dataset size: **265,487 rows**

This sampling method helps preserve hourly, daily, and monthly demand patterns while keeping the dataset manageable.

3. Data Cleaning

3.1 Column Fixing

- Removed temporary columns such as `date` and `hour`
- Standardized column names to lowercase
- Combined duplicate `airport_fee` columns

3.2 Handling Negative Values

Negative values found in monetary columns were treated as invalid entries and removed.

After cleaning, no negative monetary values remained.

3.3 Missing and Inconsistent Values

No missing values were present after cleaning. However, **4070 rows had** `passenger_count = 0`, which is considered inconsistent for completed trips.

3.4 Outlier Treatment

The following records were removed:

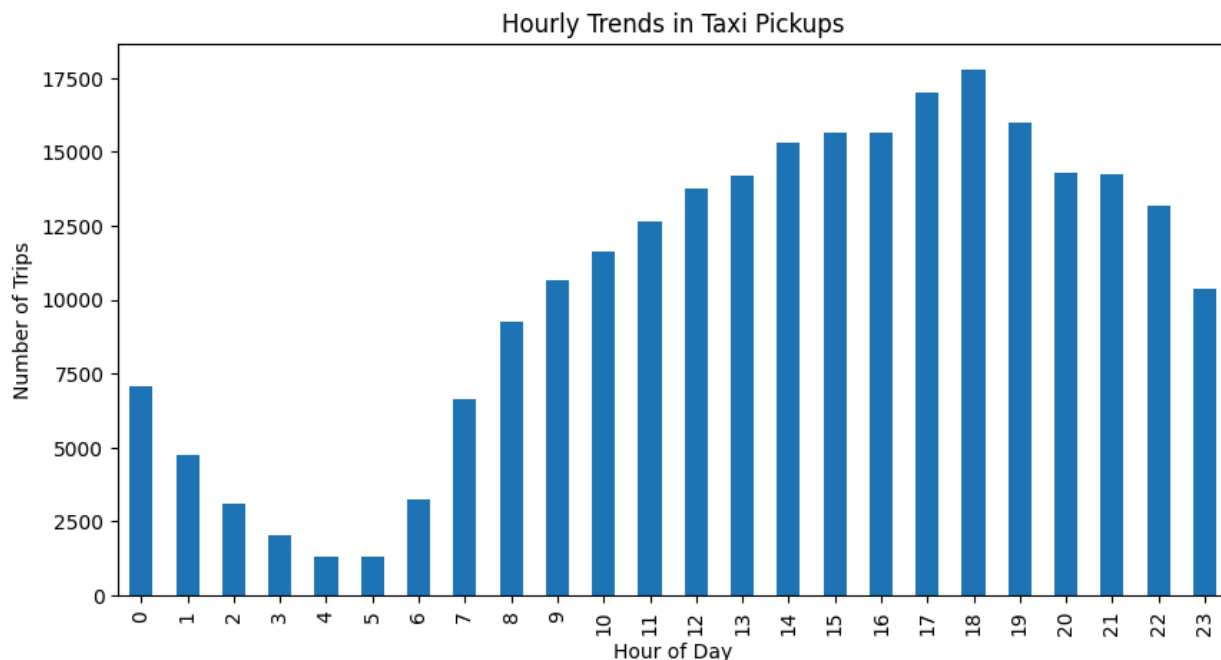
- Trips with distance < 0.1 and fare > 300
- Trips with distance > 250 miles
- Trips with fare > 300
- Invalid zero-distance trips across different locations
- Passenger count greater than 6

Final cleaned dataset contained approximately **256,000 rows**.

4. Exploratory Data Analysis

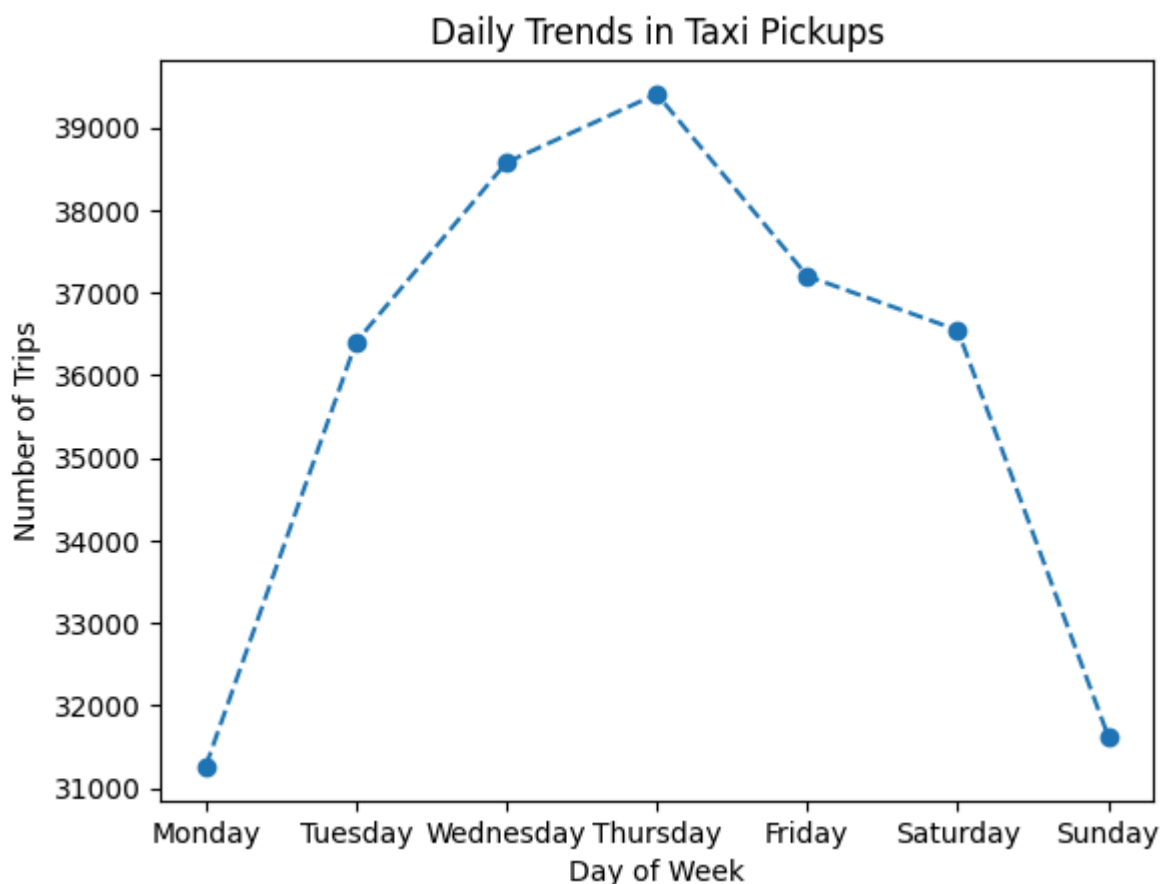
4.1 Temporal Analysis

Hourly Trends



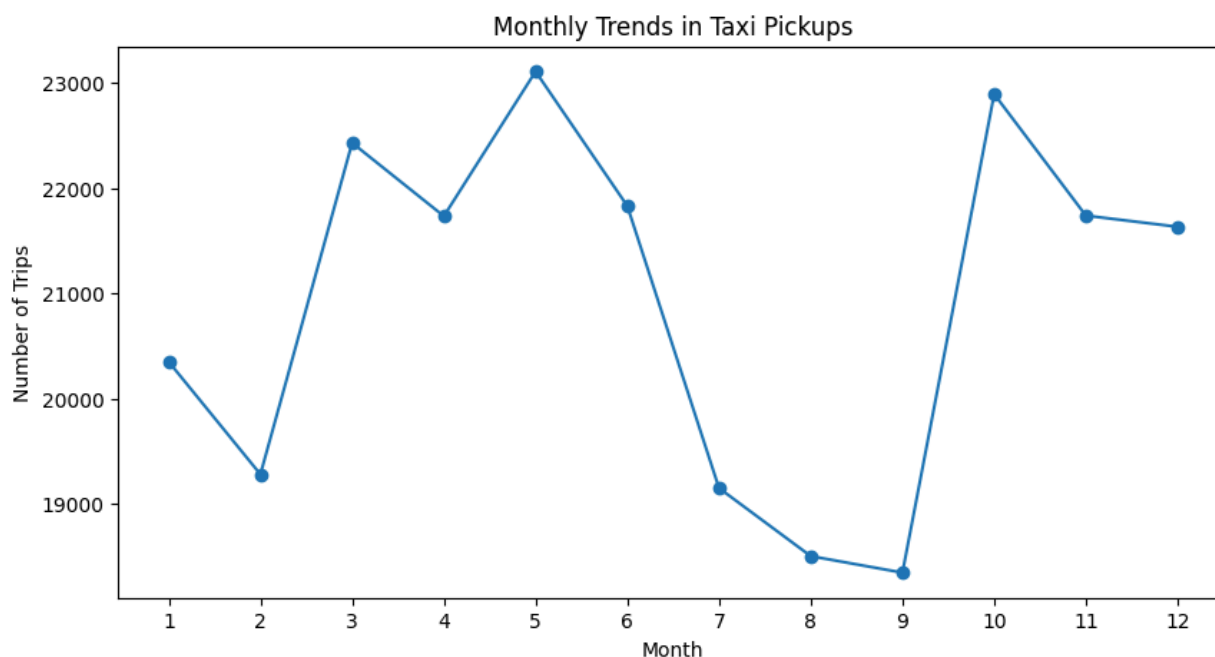
Peak demand occurs between **5 PM and 7 PM**, while the lowest demand is observed between **2 AM and 5 AM**. This suggests that evening commute hours generate the highest taxi demand.

Daily Trends



The highest number of trips occurred on **Thursday**, followed by Friday and Wednesday. The lowest trip counts were observed on Sunday and Monday, suggesting stronger weekday demand.

Monthly Trends



Taxi pickups were highest in **May and October**, while August and September had comparatively lower activity. This indicates seasonal variation in taxi usage.

4.2 Financial Analysis

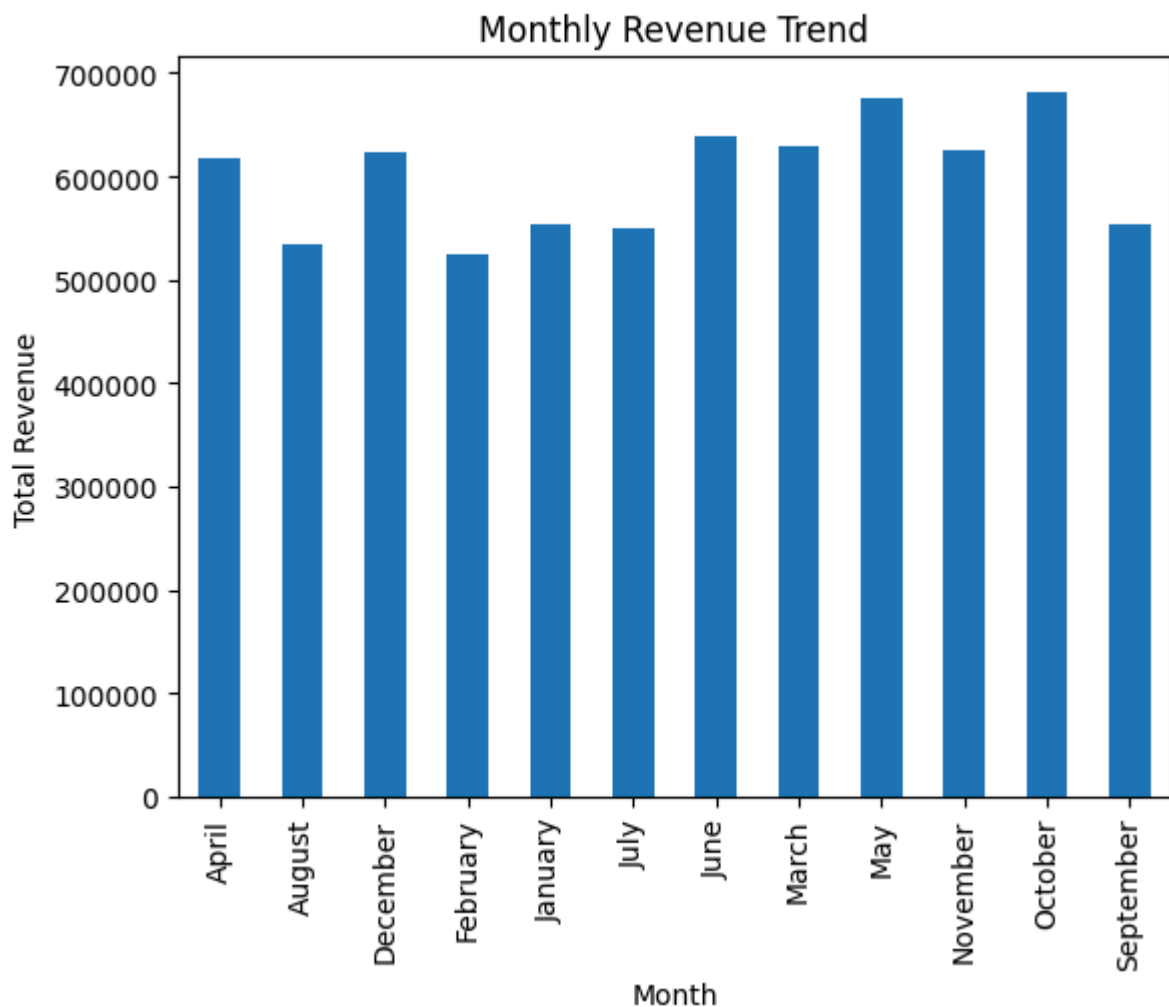
Zero values were identified in some financial and distance-related columns.

Column	Zero Count
fare_amount	69
total_amount	28
tip_amount	54821
trip_distance	2910

Zero values in `fare_amount` and `total_amount` were treated as invalid and removed. Zero values in `tip_amount` were retained because they represent trips where passengers gave no tip. Zero trip distance was not blindly removed because it can occur when pickup and drop-off locations are the same.

4.3 Revenue Analysis

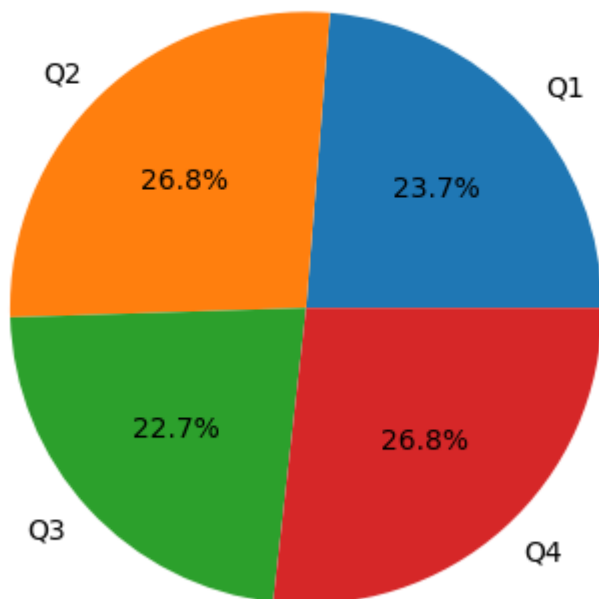
Monthly Revenue



The highest monthly revenue was observed in **October**, followed by May. The lowest revenue was observed in **February**. This suggests that revenue follows similar seasonal patterns as trip demand.

Quarterly Revenue Distribution

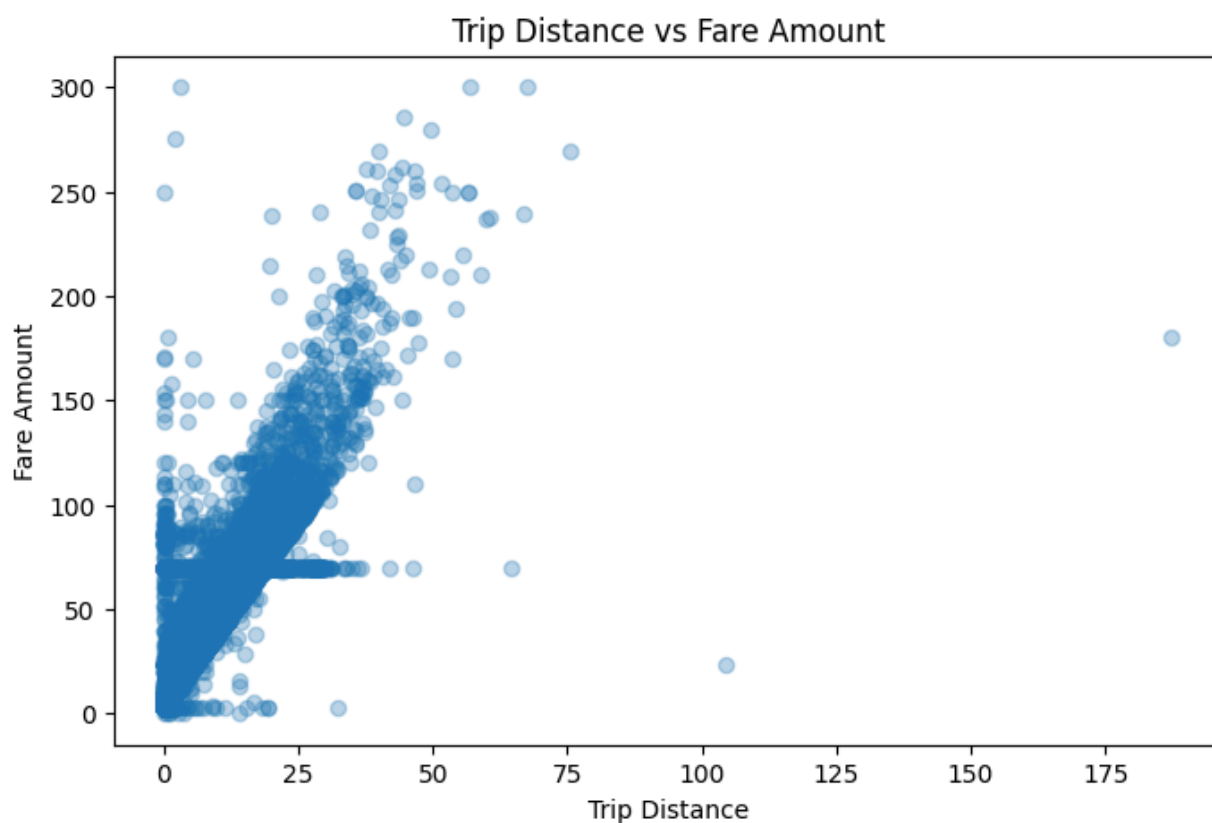
Revenue Proportion by Quarter



Revenue was distributed fairly evenly across the four quarters, with slightly higher proportions in Q2 and Q4.

4.4 Correlation Analysis

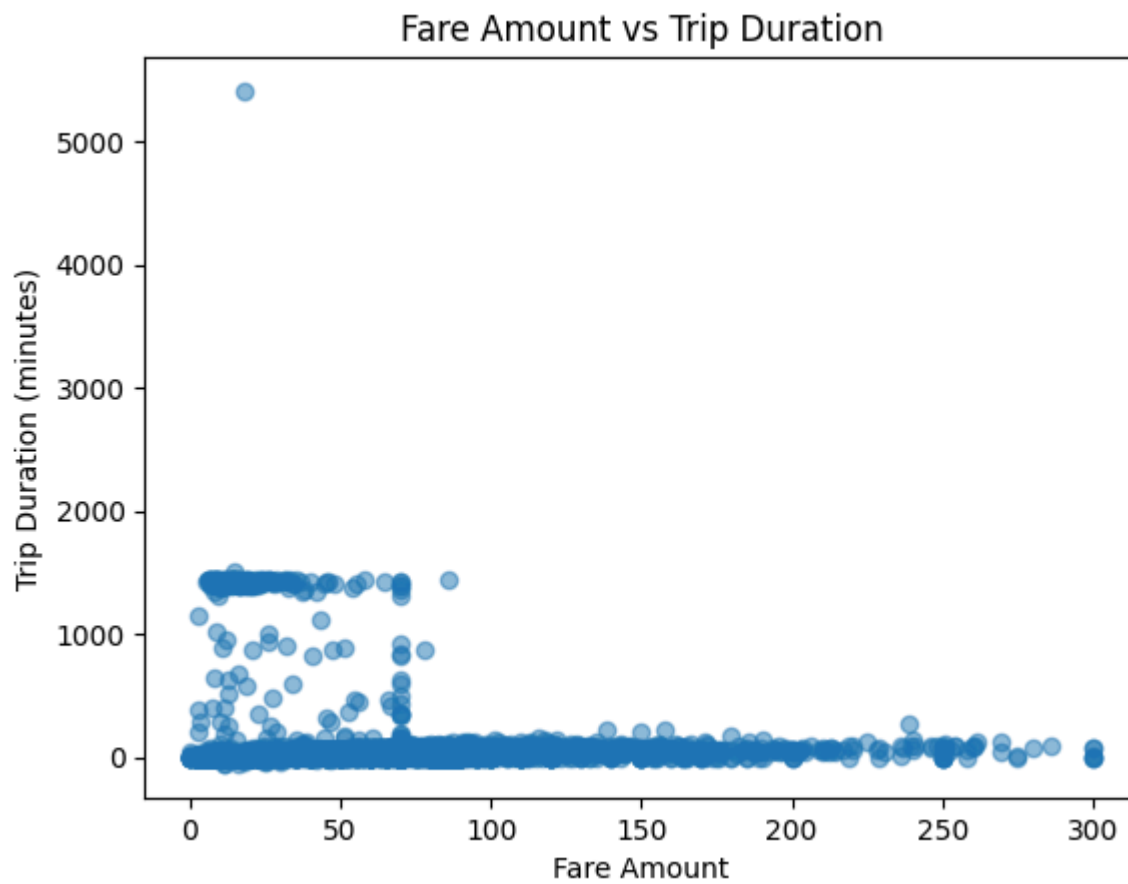
Trip Distance vs Fare Amount



The correlation between trip distance and fare amount was **0.94**, showing a strong

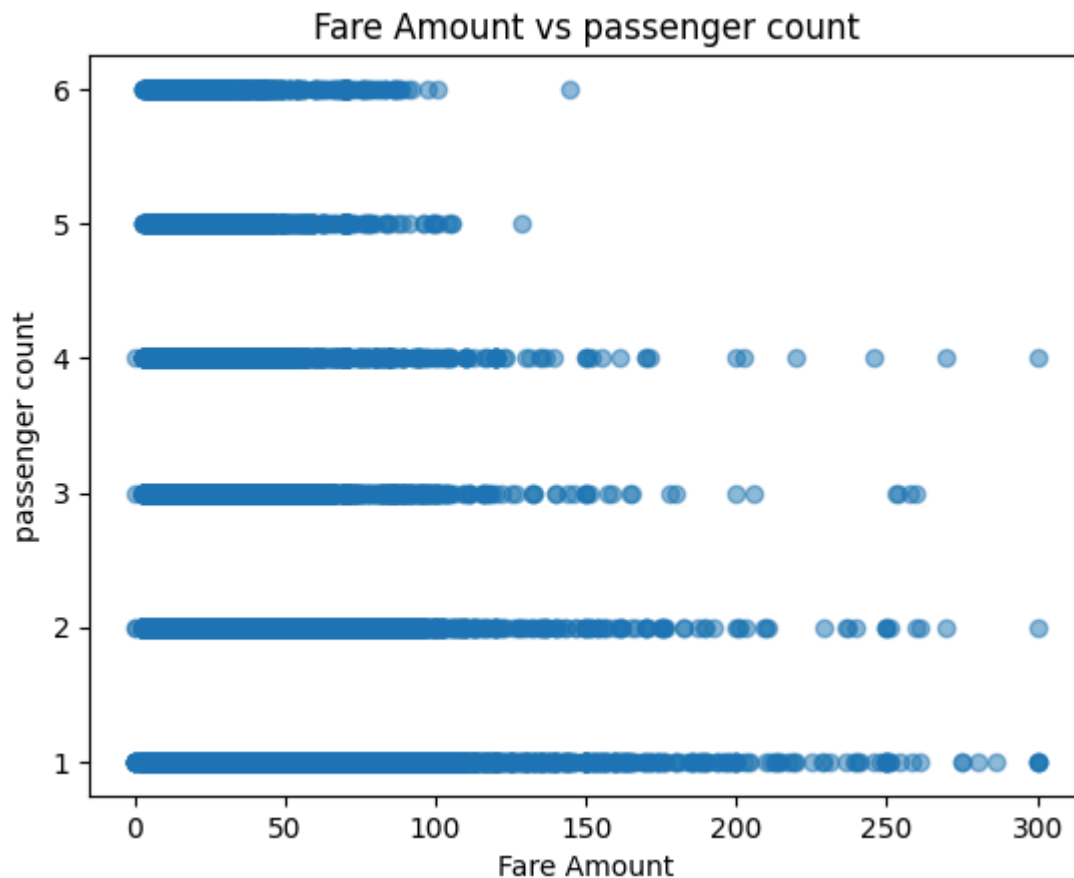
positive relationship. This means longer trips generally generate higher fares.

Fare Amount vs Trip Duration



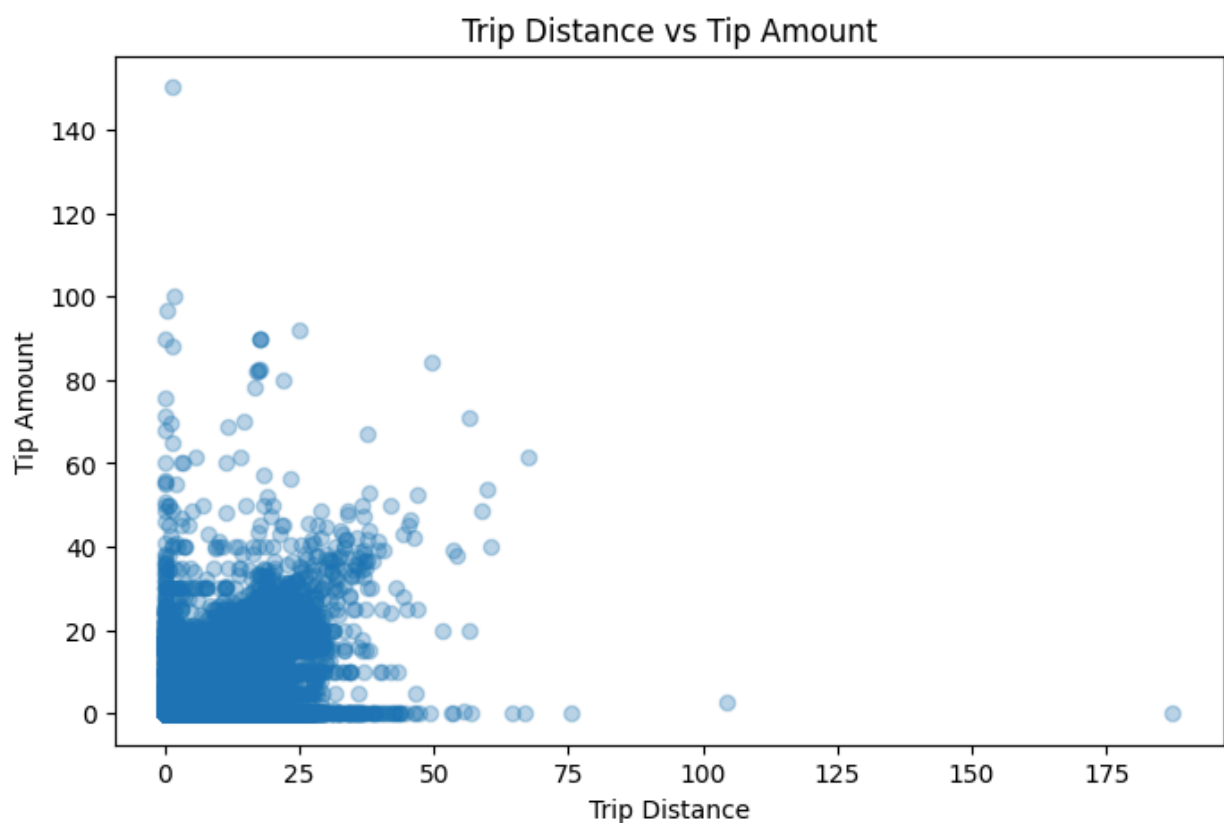
The correlation between fare amount and trip duration was **0.26**, indicating a moderate relationship. Duration affects fare, but distance appears to be a stronger factor.

Fare Amount vs Passenger Count



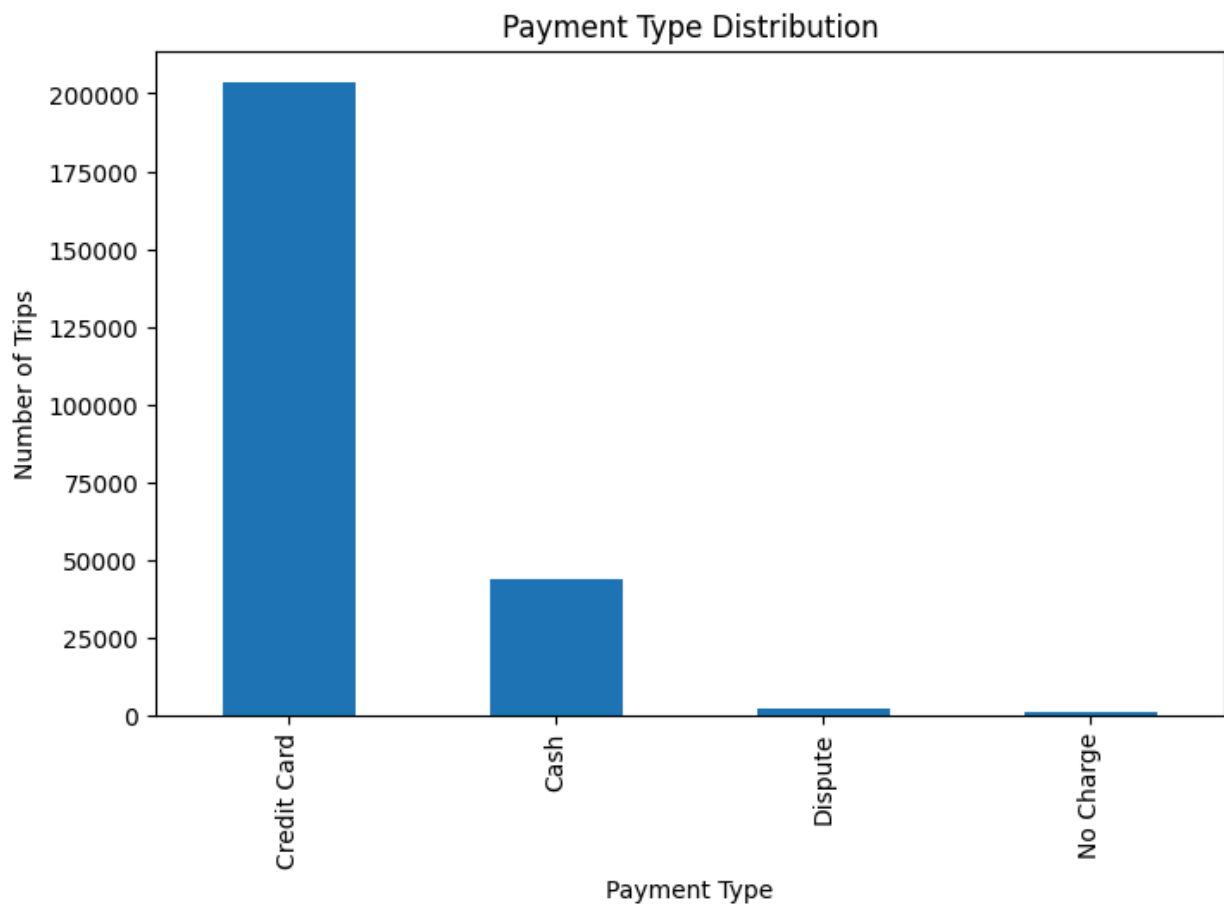
The correlation between fare amount and passenger count was **0.047**, showing a very weak relationship. This suggests that fare amount does not depend much on passenger count.

Tip Amount vs Trip Distance



The correlation between tip amount and trip distance was **0.59**, showing a moderate positive relationship. Longer trips generally tend to receive higher tips.

4.5 Payment Analysis



Most trips were paid using **credit cards**, while cash payments were lower. This indicates that digital/card payments dominate NYC taxi transactions.
